

The Effect of Strabismus and Anisometropia on the Development of Amblyopia in Synaptic Plasticity Models

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Abstract

In this work we explore the development of amblyopia from the combination of strabismus (eye misalignment) and anisometropia (unequal refractive differences) with the process of synaptic plasticity. We introduce a natural-image binocular environment with asymmetry between the channels produced by refractive blurring, eye misalignment and eye jitter where we can compare normal visual development to deprived development. The cortical responses are simulated using a computational model of neural plasticity, the Bienenstock, Cooper, and Munro (BCM) model¹. Simulations are performed to explore ocular dominance changes in normal development, development with strabismic and anisometropic deficits as well as treatment with corrective optics. The results show a remarkable robustness of ocular dominance to strabismic deficits where anisometropia produces strong ocular dominance shifts.

Significance Statement Paste your significance statement here. Please note that it should not exceed 120 words, but should be at least 50 words in length. It should not include any references.

Main Text

Introduction

Some review of amblyopia.²

Results

Paste your results here.

Discussion

Paste your discussion here.

30 **Materials and Methods**

31 Paste your materials and methods section here.

32 **Acknowledgments**

33 Paste your acknowledgments here.

34 **References**

- 35 1. Bienenstock, E.L., Cooper, L.N., and Munro, P.W. (1982). Theory for the development of neuron selectivity: Orientation specificity and binocular interaction in visual cortex. *Journal of Neuroscience* 2, 32–48.
- 36 2. Thompson, B. (2021). Neural Plasticity in Amblyopia. In *Oxford Research Encyclopedia of Psychology* (Oxford University Press). <https://doi.org/10.1093/acrefore/9780190236557.013.702>.

37 **Figures and Tables**

38 **Figure 1.**

39 **Figure 2.** Type or paste legend here. Paste figure above the legend.

40 **Table 1.** Type or paste table title here. Paste table below the title.

41 **Table 2.** Type or paste table title here. Paste table below the title.